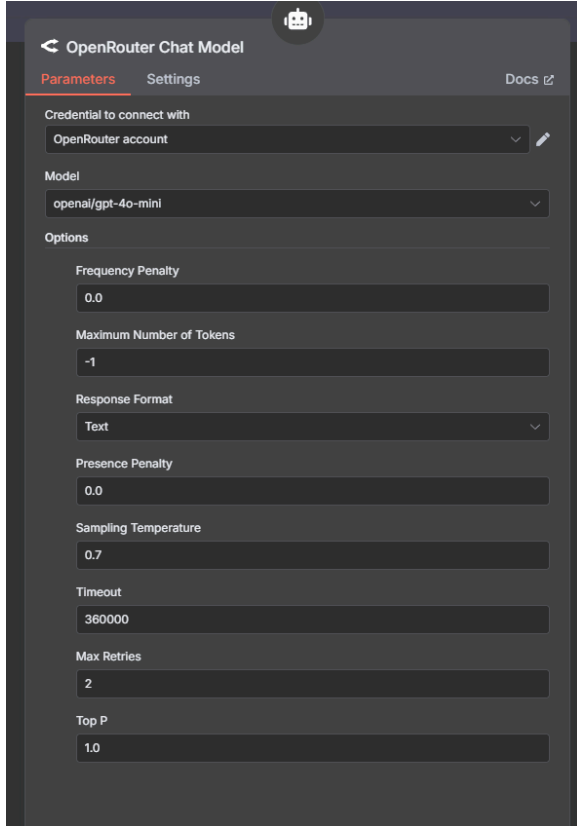


Explanation of AI Model Options

Here's what each option in this screenshot does, along with examples of when you might want to change them:



The screenshot shows the 'OpenRouter Chat Model' settings page. It has a dark theme and a sidebar with 'Parameters' (selected), 'Settings', and 'Docs' links. The main content area is divided into sections: 'Credential to connect with' (set to 'OpenRouter account'), 'Model' (set to 'openai/gpt-4o-mini'), and 'Options'. The 'Options' section contains several sliders and dropdowns: 'Frequency Penalty' (0.0), 'Maximum Number of Tokens' (-1), 'Response Format' (Text), 'Presence Penalty' (0.0), 'Sampling Temperature' (0.7), 'Timeout' (360000), 'Max Retries' (2), and 'Top P' (1.0).

Frequency Penalty

- What it does: Discourages the model from repeating the same words or phrases. Higher values (up to 2.0) make repetition less likely, while negative values encourage repetition [12](#).
- When to change: If your outputs are too repetitive (e.g., "very very very important"), increase the penalty. For poetry or technical writing where repetition is sometimes needed, use a lower or negative value [2](#).

Maximum Number of Tokens

- What it does: Sets the maximum length (in tokens) for the model's response. A token is roughly 4 characters or $\frac{3}{4}$ of a word [34](#).
- When to change: If you want shorter, more concise answers, set a lower limit. For long-form content (e.g., essays, reports), increase the limit. If set to -1, it may indicate no explicit limit, but actual limits depend on the model's architecture [34](#).

Response Format

- What it does: Specifies the format of the output (e.g., plain text, JSON, etc.).
- When to change: Choose "Text" for natural language responses, or another format if you need structured data for further processing.

Presence Penalty

- What it does: Penalizes the model for using any word it has already used, even once. This encourages more diverse word choices [5](#).
- When to change: Increase if you want highly varied language (e.g., brainstorming, creative writing). Decrease for technical or repetitive tasks where consistency is important [5](#).

Sampling Temperature

- What it does: Controls randomness in the output. Lower values (<1) make responses more predictable and focused; higher values (>1) increase creativity and randomness [67](#).
- When to change: Use a low temperature (e.g., 0.2–0.5) for factual, precise answers. Use a higher temperature (e.g., 0.8–1.2) for creative writing or brainstorming [67](#).

Timeout

- What it does: Sets the maximum time (in milliseconds) the system will wait for a response before giving up [89](#).
- When to change: Increase if you expect slow responses (e.g., complex queries or heavy server load). Decrease for applications needing fast feedback, like chatbots [89](#).

Max Retries

- What it does: Number of times the system will retry a failed request before giving up.
- When to change: Increase if you want more robustness against temporary errors. Decrease if you want faster failure feedback.

Top P

- What it does: Controls the diversity of the output by only considering the smallest set of most probable next words whose cumulative probability exceeds the value of p (typically 0.9–1.0) [6](#).

- When to change: Lower Top P (e.g., 0.8) for more focused, less diverse outputs. Higher Top P (e.g., 1.0) for more varied and creative responses [6](#).
 - NOTE: It is generally recommended to alter the temperature or the top P, but not both.
-

Summary Table

Option	What It Controls	Example Use Case for Adjustment
Frequency Penalty	Repetition of words/phrases	Increase to reduce loops in chatbot answers
Max Number of Tokens	Length of output	Increase for essays, decrease for summaries
Response Format	Output type (text, JSON, etc.)	Use JSON for API integration, Text for chat
Presence Penalty	Use of unique words	Increase for brainstorming, decrease for code
Sampling Temperature	Randomness/creativity	Higher for stories, lower for technical answers

Timeout	Wait time for response	Increase for slow servers, decrease for chatbots
Max Retries	Error recovery attempts	Increase for unreliable networks
Top P	Diversity of next-word choices	Lower for focused output, higher for creativity

Example Scenario

Suppose you're building a creative writing assistant:

- Increase Sampling Temperature to 1.0 for more imaginative text.
- Increase Top P to 0.95 for diverse word choices.
- Increase Presence Penalty to encourage unique vocabulary.
- Set a higher Max Number of Tokens for longer stories.

For a technical Q&A bot:

- Lower Sampling Temperature to 0.3 for accuracy.
- Lower Top P to 0.8 for focused answers.
- Keep Frequency and Presence Penalties low to maintain clarity and consistency.

Adjusting these parameters allows you to tailor the AI's behavior to your specific application needs.